Pin Out Table

Stepper Carriage Positioning — Micro850 PLC | 24 VDC System

Pin #	Group	Device / Signal	Wire Function	Voltage
1	Stepper Motor	Stepper Motor (STP-MTR-23055, NEMA 23)	A+ (RED) → CW250 A+	—
2	Stepper Motor	Stepper Motor (STP-MTR-23055, NEMA 23)	A− (WHT) → CW250 A−	—
3	Stepper Motor	Stepper Motor (STP-MTR-23055, NEMA 23)	B+ (GRN) → CW250 B+	—
4	Stepper Motor	Stepper Motor (STP-MTR-23055, NEMA 23)	B− (BLK) → CW250 B−	—
5	Stepper Driver	Stepper Driver (CW250/2/8)	VCC ← Driver power supply +	20–60 VDC
6	Stepper Driver	Stepper Driver (CW250/2/8)	GND ← Driver power supply −	Driver 0 V
7	Stepper Driver	Stepper Driver (CW250/2/8)	CP+ (pulse) ← PLC DO_01 (PTO)	24 VDC
8	Stepper Driver	Stepper Driver (CW250/2/8)	CP− → −24 VDC common	−24 VDC
9	Stepper Driver	Stepper Driver (CW250/2/8)	CW+ (direction) ← PLC PTO direction output	24 VDC
10	Stepper Driver	Stepper Driver (CW250/2/8)	CW− → −24 VDC common	−24 VDC
11	Stepper Driver	Stepper Driver (CW250/2/8)	H/D (½ / ⅛ step select) ← PLC DO_06	24 VDC
12	Green Pushbutton	Pushbutton, Green, 1 N.O. (GCX1102) — Left	N.O. contact → PLC DI_07	24 VDC
13	Green Pushbutton	Pushbutton, Green, 1 N.O. (GCX1102) — Left	N.O. contact ← +24 VDC	+24 VDC
14	Red Pushbutton	Pushbutton, Red (GCX1101 + ECX1040 N.O.) — Right	N.O. contact → PLC DI_02	24 VDC
15	Red Pushbutton	Pushbutton, Red (GCX1101 + ECX1040 N.O.) — Right	N.O. contact ← +24 VDC	+24 VDC
16	Yellow Pushbutton	Pushbutton, Yellow, 1 N.O. (GCX1103) — Center	N.O. contact → PLC DI_04	24 VDC
17	Yellow Pushbutton	Pushbutton, Yellow, 1 N.O. (GCX1103) — Center	N.O. contact ← +24 VDC	+24 VDC
18	Enable Switch	Selector, 2-pos maintained, 1 N.O. (GCX1350) — Enable	N.O. contact → PLC DI_03	24 VDC
19	Enable Switch	Selector, 2-pos maintained, 1 N.O. (GCX1350) — Enable	N.O. contact ← +24 VDC	+24 VDC
20	Limit Switch 1	Micro Limit Switch (WS1-Z5-W300R300) — Left / Home	COM ← +24 VDC	+24 VDC
21	Limit Switch 1	Micro Limit Switch (WS1-Z5-W300R300) — Left / Home	NO → PLC DI_05	24 VDC
22	Limit Switch 1	Micro Limit Switch (WS1-Z5-W300R300) — Left / Home	NC — not used (insulate)	—
23	Limit Switch 2	Micro Limit Switch (WS1-Z5-W300R300) — Right	COM ← +24 VDC	+24 VDC

24	Limit Switch 2	Micro Limit Switch (WS1-Z5-W300R300) — Right	NO → PLC DI_06	24 VDC
25	Limit Switch 2	Micro Limit Switch (WS1-Z5-W300R300) — Right	NC — not used (insulate)	—
26	Proximity Sensor 1	Inductive Prox. (PNM6-AP-3A) — Left-most	Signal wire (BK) → PLC DI_09	24 VDC
27	Proximity Sensor 1	Inductive Prox. (PNM6-AP-3A) — Left-most	Power −24 VDC (BU)	−24 VDC
28	Proximity Sensor 1	Inductive Prox. (PNM6-AP-3A) — Left-most	Power +24 VDC (BN)	+24 VDC
29	Proximity Sensor 2	Inductive Prox. (PNM6-AP-3A) — Center	Signal wire (BK) → PLC DI_08	24 VDC
30	Proximity Sensor 2	Inductive Prox. (PNM6-AP-3A) — Center	Power −24 VDC (BU)	−24 VDC
31	Proximity Sensor 2	Inductive Prox. (PNM6-AP-3A) — Center	Power +24 VDC (BN)	+24 VDC
32	Proximity Sensor 3	Inductive Prox. (PNM6-AP-3A) — Right-most	Signal wire (BK) → PLC DI_10	24 VDC
33	Proximity Sensor 3	Inductive Prox. (PNM6-AP-3A) — Right-most	Power −24 VDC (BU)	−24 VDC
34	Proximity Sensor 3	Inductive Prox. (PNM6-AP-3A) — Right-most	Power +24 VDC (BN)	+24 VDC
35	Green Status Light	LED Pilot Light, Green (GCX1232-24L) — Left	Lamp + ← PLC DO_07	+24 VDC
36	Green Status Light	LED Pilot Light, Green (GCX1232-24L) — Left	Lamp − → −24 VDC common	−24 VDC
37	Red Status Light	LED Pilot Light, Red (GCX1231-24L) — Right	Lamp + ← PLC DO_08	+24 VDC
38	Red Status Light	LED Pilot Light, Red (GCX1231-24L) — Right	Lamp − → −24 VDC common	−24 VDC
39	Yellow Status Light	LED Pilot Light, Yellow (GCX1233-24L) — Center	Lamp + ← PLC DO_09	+24 VDC
40	Yellow Status Light	LED Pilot Light, Yellow (GCX1233-24L) — Center	Lamp − → −24 VDC common	−24 VDC
41	HMI Display	PanelView 800 4.3" TFT HMI (2711R-T4T)	Power +24 VDC	+24 VDC
42	HMI Display	PanelView 800 4.3" TFT HMI (2711R-T4T)	Power −24 VDC	−24 VDC
43	HMI Display	PanelView 800 4.3" TFT HMI (2711R-T4T)	Ethernet ↔ Micro850 Ethernet port	—
44	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	+24 VDC ← PSB24-060S +V	+24 VDC
45	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	−24 VDC ← PSB24-060S −V	−24 VDC
46	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	DI COM → −24 VDC (sinking inputs for PNP devices)	−24 VDC
47	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	DO +CM ← +24 VDC	+24 VDC
48	PLC Power & Commons	Allen-Bradley Micro850 (2080-LC50-24QBB)	DO −CM → −24 VDC	−24 VDC

Color Legend

Stepper Motor — STP-MTR-23055 (coil leads A± / B±)
Stepper Driver — CW250/2/8 (power, CP / CW pulse-direction, H/D)
Operator Controls (GCX 22 mm pushbuttons, enable selector)
Limit Switches 1 & 2 — WS1-Z5-W300R300 (left / home, right)
Proximity Sensor 1 — Left-most
Proximity Sensor 2 — Center
Proximity Sensor 3 — Right-most
Status Lights — GCX 24 V LED pilot lights
PanelView 800 4.3" TFT touch screen HMI
Micro850 PLC power and I/O commons

Notes

CW250 inputs: confirm CP / CW / H/D accept 24 V signals; if rated 5 V, add a series resistor per the driver manual.

Set CW250 phase current to the motor rating (2.8 A/phase for STP-MTR-23055). H/D assumed to select ½ / ⅛ step ("/2/8").

PTO direction output is set in CCW Motion › X_axis; wire CW+ to that output.

GCX1101 ships with 1 N.C.; fit an ECX1040 1 N.O. block so the red button matches the program.

Lead colors from the motor and sensor datasheets; confirm against the parts received.